

BELY, Russian FederationWMO: **265850**Lat: **55.833N**Lon: **32.933E**Elev: **222**StdP: **98.69**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-20.0	-26.1	0.4	-23.3	-22.8	0.5	-19.6	8.6	-5.4	7.9	-3.4	1.3	90	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	27.7	19.2	26.0	18.8	24.4	17.9	20.7	25.5	19.6	24.3	18.6	23.1	2.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.0	23.6	17.8	13.1	22.0	16.9	12.4	21.0	60.5	25.4	56.8	24.1	53.6	23.2	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.4	5.8	-28.1	29.9	2.7	1.7	-30.0	31.1	-31.6	32.0	-33.1	33.0	-35.0	34.2		
(5)			-28.2	21.7	2.6	1.1	-30.1	22.4	-31.7	23.1	-33.2	23.7	-35.1	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-6.9	-5.8	-1.5	6.6	11.7	16.1	17.9	15.7	10.5	5.2	-0.8	-4.7		
	DBStd	9.96	6.11	6.51	4.60	5.23	4.89	3.72	3.08	3.08	3.75	4.23	5.44	6.33		
	HDD10.0	2465	523	442	357	131	38	2	0	0	38	155	323	456		
	HDD18.3	4792	781	676	616	353	214	86	46	94	234	406	573	714		
	CDD10.0	784	0	0	0	28	89	185	244	176	54	7	0	0		
	CDD18.3	70	0	0	0	0	7	20	32	11	0	0	0	0		
	CDH23.3	571	0	0	0	2	56	154	257	95	7	0	0	0		
	CDH26.7	91	0	0	0	0	6	21	54	10	0	0	0	0		
(14) Wind	WSAvg	2.5	2.8	2.9	2.9	2.3	2.3	2.0	1.8	1.9	2.1	2.8	2.9	3.0		
(15) Precipitation	PrecAvg	645	44	36	33	31	60	77	87	72	72	58	49	41		
	PrecMax	825	66	64	64	52	101	174	187	132	122	142	101	63		
	PrecMin	451	12	12	10	10	23	15	16	1	36	7	1	18		
	PrecStd	120	12	13	14	14	24	40	54	34	27	34	26	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	4.7	12.6	23.2	27.4	28.8	30.6	28.1	24.2	19.7	10.7	7.5		
		MCWB	2.6	2.7	7.3	13.4	18.0	19.0	21.3	19.6	18.4	14.3	9.6	6.6		
	2%	DB	2.0	3.0	8.9	20.7	25.1	27.0	28.3	25.9	20.9	14.5	8.9	4.8		
		MCWB	1.3	1.4	4.2	12.5	16.8	18.2	20.2	18.8	15.6	11.4	7.8	4.0		
	5%	DB	1.2	2.0	6.3	18.3	22.4	25.1	26.2	24.0	18.5	12.5	7.7	3.1		
		MCWB	0.5	0.7	2.9	11.1	15.4	18.2	19.5	18.1	14.1	10.4	6.8	2.5		
	10%	DB	0.5	1.3	4.3	15.5	19.8	23.1	24.3	22.3	16.4	10.9	6.0	1.9		
		MCWB	-0.1	0.3	1.9	10.0	14.0	17.1	18.2	17.2	13.0	9.3	5.2	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.0	7.7	14.6	19.2	20.9	22.7	21.2	18.9	14.5	9.9	6.8		
		MCDB	3.1	4.3	11.7	21.7	25.2	26.6	28.6	25.8	21.8	19.3	10.4	7.5		
	2%	WB	1.6	1.7	4.9	12.7	17.8	19.7	21.3	19.7	16.3	12.2	8.1	4.0		
		MCDB	1.9	2.5	7.4	19.0	23.4	25.1	26.5	24.2	20.0	13.8	8.7	4.7		
	5%	WB	0.7	1.1	3.5	11.6	16.4	18.6	20.0	18.7	14.7	10.7	6.7	2.6		
		MCDB	1.0	1.7	5.5	17.1	20.7	23.2	24.5	23.1	17.2	12.1	7.5	3.0		
	10%	WB	0.0	0.4	2.2	10.4	14.9	17.8	18.9	17.6	13.5	9.6	5.3	1.3		
		MCDB	0.5	1.1	4.1	15.0	18.8	22.0	23.3	21.6	15.9	10.9	6.0	1.8		
(35) Mean Daily Temperature Range	MDBR		4.7	6.0	7.0	8.8	9.8	9.1	9.1	9.2	8.2	6.0	4.2	4.0		
	5% DB	MCDBR	3.2	4.2	9.7	13.3	12.9	11.9	12.4	12.7	11.8	8.6	4.9	3.5		
		MCWBR	3.3	3.6	6.1	6.7	6.4	5.3	5.5	6.7	7.3	5.9	4.4	3.3		
	5% WB	MCDBR	3.1	3.5	7.8	11.8	11.8	10.4	10.7	11.9	10.3	7.2	4.7	3.4		
		MCWBR	3.3	3.3	5.4	6.4	6.6	5.2	5.4	7.0	7.0	5.3	4.4	3.2		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.318	0.340	0.398	0.387	0.368	0.397	0.411	0.363	0.344	0.320	0.282		
	taud		2.143	2.105	2.163	2.200	2.334	2.405	2.336	2.290	2.398	2.393	2.257	2.155		
	Ebn at Noon		605	724	813	816	854	875	838	797	791	708	563	509		
	Edh at Noon		60	91	110	124	116	109	115	113	88	70	55	47		
(44) All-Sky Solar Radiation	RadAvg		0.57	1.42	2.93	3.99	5.18	5.46	5.33	4.25	2.77	1.27	0.47	0.34		
	RadStd		0.12	0.21	0.29	0.40	0.44	0.52	0.53	0.50	0.29	0.26	0.09	0.10		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	N/A

Nomenclature: See separate page